1 General

In this prototype Communication Design is prioritized over other types of design. Thus the prototype will focus on communicating the general actions and mechanics that the player can do as well as how to do those actions and mechanics. This is done through a tutorial where the player will be introduced to the mechanics of the game. After the tutorial the player will then be put into a series of levels where they will use the mechanics they learnt in the tutorial. During both the tutorial and levels important game elements will be highlighted in order to get the player to recognize them quicker and help the player complete the game.

2 Mechanics

What can the player do:

1. Move around with WASD.
2. Sprint using Shift and WASD
3. Interact with the environment through Space.
4. Toggling the player's light with R.

Why can the player do these options:

1. Explore the map
2. Run away from enemies or map hazards
3. To pick up keys, open doors, hide in hiding spots, switching lights on, flipping levers and solving puzzles.
4. To see places that can only be seen when the light is off or to hide from an enemy.

How will these mechanics be balanced

1. The player speed will be slower than the base enemy speed but still fast enough to not make moving around feel like a drag.
2. The player sprint speed will be faster than base enemy speed but the player will only be able to sprint for a set amount of time.
3. The player will only be able to interact with one interaction at a time if the player is carrying an item and wants to open a door the player will first have to drop the item before opening the door.
4. The player will not be able to see the map clearly when their light is turned off, thus the player will need to toggle between light on and light off states.

How will these mechanics be implemented.

1. A player script will move the player.
2. A sprint script will increase the player's speed and keep track of how long the player has been sprinting
3. Two scripts will be made one which defines the interaction and another which allows the player to interact with the defined interaction.
4. A script will read player input and if they press the correct button ,R, the light will switch off.

3 Challenges

The player can use the mechanics in order to beat some challenges from the game. These challenges include running away from the enemy by using the sprint mechanic or hiding from the enemy by either finding a hiding spot and turning off the player light or by interacting with a game element hiding spot. The player will also need to move items from one space to another and they may have to drop the item in order to hide from enemies which causes another challenge. The player would need to remember where they dropped their item due to the player being able to accidentally throw their item.

4 Hypothesis

If important game elements are highlighted then players will recognize that element faster because the information hierarchy directs the players attention.

5 Goals

Some goals for the prototype

1. To create a comprehensive tutorial in which the player learns all the game mechanics and uses them in order to understand the mechanics in such a way that they do not need to constantly be reminded of them.
2. To create a user interface that can communicate the current state the player is in.
3. To make specific code from the first prototype that can only be used in specific situations more general so that the code can be reused easily and is scalable for new features.

6 Genre

The main genre is Puzzle. The prototype is of the puzzle genre because the player needs to interact with different interactables in order to progress. The way the player needs to interact is something that the player needs to figure out. The sub genre is Exploration. The prototype is of the exploration genre because the player needs to explore the map they were put in to progress with the puzzle they are faced with.

6.1 Game Examples

Portal 1 and Portal 2

In both Portal 1 and Portal 2 the player is given a portal gun which allows the player to shoot two portals and walk through them . The player can also pick up items with the portal gun. In both games you can only have two portals out at a time this leads the player to think about how they are going to use their portals because of its limited use. It is also communicated where the player can shoot their portals with a white surface, this communication also helps the player in how they should approach the puzzle they are presented with. Using the portals the player also needs to explore the levels to find how they should use their portals to beat the level. In the prototype only being able to interact with one interactable at a time will have a similar effect to the limited use of the portal gun in the Portal games. This can lead to situations where the player has to hide from an enemy but they are currently carrying an item leading the player to drop the item away from the hiding spot and then hiding before the enemy catches them.

7 Communication Design

The main mechanics of the prototype will be shown to the player through a tutorial.

The moving mechanic will be shown by showing the buttons on the screen while requiring the player to use them to move through the map. Where the player needs to move is shown through a light.

The sprinting mechanic will be shown by showing the button needed to sprint on screen and giving the player a long part of the map in which they need to sprint to get through. If the player takes too long getting through this part they will need to try it again.

The sprinting mechanic cannot be used forever thus a stamina bar will be shown on screen which will drain to empty if the player sprints for too long. This is to show the player that they cannot sprint forever and that they need to be mindful of where and for how long they sprint.

The interact mechanic will be shown to the player by showing different interactions they can interact with while highlighting the important intractables

in different ways. The player will only be able to interact with one interactable at a time.

The game will show the player that they can press R on the keyboard to turn off their light. It will also show an indication in areas where you need to turn off the light in order to progress. This indication is a purple diamond in the world space near the hidden objects. When an enemy is nearby and you need to turn off the light the light will change colour. The colour chosen is a yellow warning colour for enemies.

Different ways to bring attention to game elements

- Shine a light on the important element. This way works because the game is dark and the player only has limited vision of what is around them.
- Change the element's colour to make it stand out from the background.
- Put instructions in the foreground to bring the element to the player's attention.

8 Process

This prototype will be a continuation from the previous prototype. Thus scripts and assets from the previous prototype will be used. Although it is a continuation a new map was made because the new map shows off more of the Hypothesis and communication design which is highlighting important game elements to allow the player to identify them faster. The same tilemap was used from the first prototype.

The Cinemachine package was installed from the unity registry in order to do the camera work. Cinemachine is a unity package made by unity to give developers more control over the unity camera. A Cinemachine camera was made to follow the player.

Added the NavMeshPlus package from GitHub [1] in order to do 2d ai navigation. This package helps takes the default unity navigation mesh, which is predominantly used for 3d projects, and allows it to work in a 2d environment. Using this package enemies can easily move around the 2d map.

A coding system where a general script is made for different general interactions while specific scripts where used with inheritance from the general script to account for the specific situations was made.

The coding system was used in the trigger system to allow for the text to change when the player reaches certain spots on the map. A subscript was made to do the sprinting tutorial.

The coding system was used in the interaction system that allowed the player to interact with different game elements. A subscript was made for each different interaction the player can do.

In the tutorial the areas between the different mechanic tutorials were extended to allow the players to read about the mechanics and not just need to

experience them. Not much reading was given in order to not overwhelm the player with new information.

The colour decisions for the highlighting lights where based on a traffic light. Green for the important elements, yellow for warning against nearby enemies and red for the enemies.

8.1 Hypothesis in prototype

In the prototype areas where the player should move are highlighted with a coloured light.

Important objects that the player would need to pickup to continue in the game are also highlighted while the area where they should be used are highlighted to show the player where they should use the items.

Enemies have a red light around them to signal danger to the player, while the player's own light turns yellow when the enemy is nearby.

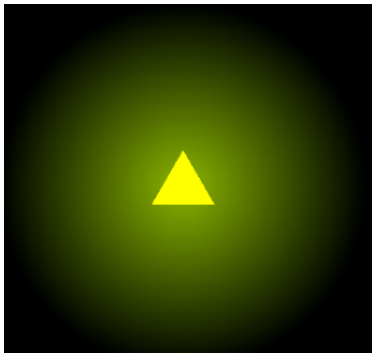

Figure 1: Image of key highlight

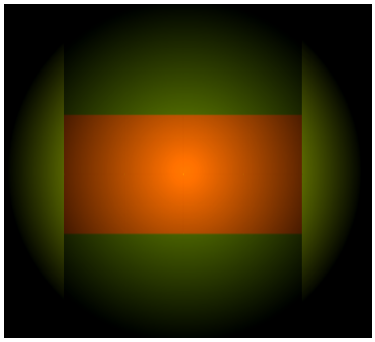

Figure 2: Image of door highlight

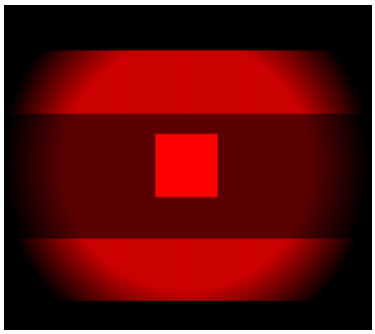

Figure 3: Image of enemy highlight

Figure 4: Image of player warning

9 Reflection

During the designing of the prototype I enjoyed changing the code from the first prototype. The code I usually changed followed the following criteria: Code that can only be used in a single specific situation. One example of this code was the item script that I made during the first prototype, I changed it to a general interaction script in this prototype which caused the efficiency of coding to go up because I reused the code for different situations. A change from the first to the second prototype was the picking up of items. In the first prototype the script was done to only work with items thus the player could only interact with items. In the second prototype it was changed to work with any interaction in general by making an interaction root script which any other type of interaction inherits from. Time management was an issue during this prototype because other assignments had to be done while doing this prototype. This issue helped me improve my time management to be able to work on all assignments by allocating them all time every day. I would say I achieved the goals I set out to do.

During playtesting no highlighting was done to see how the players would play the game. The players diamond not know where to go and were generally lost to what they should do. After adding the highlights to important game elements the players went through the tutorial much faster than when they did not have the highlight and the playtesters were also not lost during the tutorial on what they are supposed to do. This evidence suggests that the Hypothesis is correct.

10 References

[1]h8man, "NavMeshPlus," GitHub, Nov. 26, 2022. https://github.com/h8man/NavMeshPlus